

SALAR SOHRABI

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SUMMARY

Senior Embedded Software Engineer with 15+ years of experience delivering safety-critical automotive and embedded Linux platforms for production ECUs, infotainment and GNSS devices. Specialised in AUTOSAR, ISO 26262, Linux BSP and high-performance C/C++17/20 across MCU and SoC platforms. Production software delivery for OEMs including Volvo Cars, Volvo Trucks and Renault Group. Strong hardware awareness including multilayer high-speed PCB design and board bring-up.

Embedded C/C++(17/20), Python, Rust, CUDA | AUTOSAR (Classic/Adaptive), ISO 26262, ASPICE

Safety-Critical SW (ASIL-B/C/D) | Linux BSP, Yocto, U-Boot, Kernel, Drivers

MCU: AURIX TC3xx, NXP S32, Nvidia Orin | Robot Framework, HIL/SIL, Jenkins, Docker

AOSP / Embedded Android, BSP, HAL | GNSS/RTK, IoT, IPC, RTOS/QNX | License: B

EXPERIENCE

Sigma Technology (Gothenburg, Sweden) - Senior Embedded Software Engineer | Dec 2022 - Present

- Designed high-performance C++17/20 IPC framework using lock-free queues and zero-copy shared memory, reducing inter-process message latency on Nvidia Orin - Volvo Cars
- Developed safety-critical software within AUTOSAR-based systems on Nvidia Orin/Xavier, DriveOS, QNX
- Implemented IPC and UDP/TCP communication components; Robot Framework integration testing on HIL
- Ensured MISRA/Polyspace static analysis compliance; built Python tooling for log analysis and validation
- Designed CI/CD pipelines (Jenkins + Docker); mentored junior engineers and performed code reviews
- Implemented safety-related Embedded C on AURIX TC3xx for powertrain ECUs - Volvo GTT
- Designed SW architecture for safety-critical ECU subsystems per ISO 26262 (ASIL-C/B); derived and documented safety requirements with full traceability
- Applied HW safety manuals (NXP/Infineon); implemented fault detection, monitoring, safe-state handling
- Developed peripheral drivers (GTM, ADC, MPU); configured AUTOSAR stack and RTA-OS
- Unit/integration testing with Cantata and Robot Framework; debugging with Lauterbach (T32)
- Developed C/C++ components for truck simulation and multi-ECU integration; system-level validation and debugging within Linux-based environments - Volvo Trucks
- Designed HEBAT battery monitoring firmware (STM32, custom binary protocol with CRC-8) and .NET/ASP.NET backend with Blazor Server and SignalR real-time streaming - In-house R&D
- Developed battery health scoring algorithms and scenario-based load simulation with programmable switching schemes
- Full telemetry pipeline (USB -> parser -> backend -> SignalR -> browser); HIL-validated, safety-enforced Capgemini Engineering (Fundao, Portugal) - Senior Embedded Software Engineer | Dec 2020 - Dec 2022
- Performed end-to-end Android platform bring-up on embedded hardware (AOSP-based infotainment)
- Developed BSP, kernel modules and HAL layers; built HIDL/AIDL binder interfaces in C++
- Developed SELinux policies for Android services; customised Android build system (Soong, device trees)
- Embedded Linux user-space daemons: multi-threading, Sockets, IPC mechanisms, RPCs, synchronisation
- Developed automotive software using C/C++, AUTOSAR, ISO 26262 (ASIL-B/C) - Renault Group
- Safety requirements analysis and ASIL decomposition, FMEA contribution; GoogleTest, MISRA/Polyspace
- Silix co. - Startup Co-founder / Embedded Linux & BSP Lead | Feb 2016 - Dec 2020
- Built 500+ channel GNSS/RTK OEM module; led 3 engineers from architecture to mass production in one year
- Developed Embedded Linux BSP from scratch: U-Boot, kernel, device tree overlays, rootfs on Cortex-A53
- Wrote Linux drivers: Bluetooth, high-speed USB, 4G WWAN, GPS, serial; boot time <10 s via Yocto
- Hand-held GPS (<2 cm accuracy) using uBlox RTK with IMU dead-reckoning for GNSS-denied areas
- Designed multilayer high-speed PCBs (6+ layers, controlled impedance) in Altium; full HW/SW co-design
- Earlier Roles - Embedded Hardware/Software Engineer | Jun 2008 - Feb 2016
- Medaria (2014-2016): Led HW/SW for CRYO/3D-Fractional medical devices (IEC 60601); Cortex-M4, 600 V PWM; multilayer PCB design
- Boreshsazan (2012-2014): DLP/SLA 3D-Printer firmware (FPGA/MCU, sub-0.1 mm accuracy); 3-axis CNC controller; PCB design with signal integrity alongside noisy power stages
- Noavaran Moj (2008-2012): 500 W offline PSU with PFC (>90% eff., PF = 0.94); digital DC electronic load (CC/CV/CR/CP); battery chargers/analysers via EIS; led team to mass production

SKILLS

Languages: C, C++(14,17,20), Python, Rust, CUDA, C#/I.NET, Assembly; MISRA/Polyspace, Robot Framework, JS (Node.js, TS), HTML/CSS

Automotive: Classic/Adaptive AUTOSAR, MCAL, ARXML, ISO 26262 (ASIL-B/C/D), ASPICE

Safety: Functional Safety Engineering, Safety Requirements Derivation, ASIL Decomposition, Safety Analysis (FMEA/FTA), Fault Detection, Safe-State Handling, CANoe, SOME/IP

Architecture: Software Architecture, HW/SW Co-design, Distributed Systems, Real-Time Event Processing

Linux: Kernel, Rootfs (Yocto/Buildroot), U-Boot, Device Tree, Shell, SELinux

Android: AOSP, Platform Bring-up, BSP, HAL (HIDL/AIDL), Binders, SELinux

RTOS: QNX, Linux (PREEMPT RT), RTA-OS, FreeRTOS, uCOS

Tools: Cantata, Lauterbach, GDB, CMake, Bazel, UML, Design Patterns

DevOps: Jenkins, Docker, GitLab CI, containerised dev environments, Grafana

Version: Git, Gerrit, GitLab, SVN | V-model, TDD, Unit Testing, Mocks

MCU/SoC: Nvidia Orin/Xavier, AURIX TC3xx, NXP S32, STM32MP15x, i.MX6/8, AM335x, STM32F7/H7, LPC43x, ZYNQ, Artix-7/Virtex/Spartan, Cyclone, TMS320

Hardware: Altium Designer, Multilayer PCB (controlled impedance), Board Bring-up, Signal Integrity

Protocols: CAN, CANopen, LIN, UDS, SOME/IP, MODBUS, RS-485, Ethernet, PCIe, MQTT, 4G/LTE, LoRA, ZigBEE, WLAN,

BLE, SDR (HackRF/USRP)

Spoken: Fluent English | Basic Swedish | Driver License: B

PROJECTS & RESEARCH

Price Structure Engine (C++20, Python, PyTorch). Real-time event processing framework for market structure analysis. Lock-free data structures, Python bindings, ML pipelines for financial time-series.

Gesture Recognition for HMI (C++, OpenCV, OpenCL, MATLAB). Industrial machine control via stereoscopy and IR camera; 10 gesture classes on PC, SBC and ZYNQ hardware.

Smart Home IoT Controller (C++, Linux, Python, React Native). Star-topology IoT on Cortex-A53/A72; LoRA, ZigBEE, WLAN, MQTT. Led 5 engineers to mass production; 10+ installations; boot <10 s via Yocto.

Satellite Receiver/Tracker (ZYNQ, Cortex-H7, Altium). Schematic, multilayer PCB and firmware for 3-axis servo tracker. 0.1 degree accuracy, 0.05 degree/s slew rate on moving-base platform.

EDUCATION

M.Sc. Electronics Engineering, Sharif University of Technology | 2005 - 2008

B.Sc. Electronics Engineering, Iran University of Science and Technology | 2001 - 2005

ADDITIONAL

Ranked 17th in national M.Sc. entrance exam (14,000 candidates) | Top 0.5% in National University Entrance Exam (200,000 candidates) | Interests: High-performance computing, embedded AI, algorithmic systems